On the equivalence between Multi-Nested Timescales DeepONet and Maximum Entropy Principle

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February 7, 2022

A linear form for the distribution function, which results in a thermalized local Boltzmann distribution within individual bins, has been employed. This function is defined as follows

$$\mathcal{F}_b\left(\epsilon_i\right): \qquad \log\left(\frac{g_i}{n_i}\right) = \alpha_b + \beta_b \epsilon_i \qquad \forall \quad i \in \mathcal{I}_b$$

where

$$\begin{aligned} \alpha_b &= \log\left(\frac{Q_b}{n_b}\right) \\ \beta_b &= \frac{1}{k_B T_b} \end{aligned} \tag{1}$$

For now, let assume $T_b = T$.

1 Timescale 1

The mass fraction of group $\mathcal{I}_P \in \mathcal{I}$ will assume the following formulation

$$Y_P = \frac{\exp(-x_P) Q_P}{\widetilde{Q}} Y_{O_2} \qquad \forall \quad \mathcal{I}_P \in \mathcal{I}$$

where x_P is the single output of the DeepONet for *Timescale 1*.
The following partition functions can also be defined

$$\widetilde{Q} = \sum_{\mathcal{I}_P \in \mathcal{I}} \exp(-x_P) Q_P \quad , \quad Q = \sum_{\mathcal{I}_P \in \mathcal{I}} Q_P$$

and

$$Q_P = \sum_{i \in \mathcal{I}_P} q_i = \sum_{i \in \mathcal{I}_P} g_e g_i \exp\left(-\frac{\epsilon_i}{k_B T}\right) \qquad \forall \quad \mathcal{I}_P \in \mathcal{I}$$

The number density of group $\mathcal{I}_P \in \mathcal{I}$ can be expressed as follows

$$\begin{aligned} n_P &= Y_P \frac{\rho}{m} \\ &= \frac{\exp(-x_P) Q_P}{\widetilde{Q}} Y_{O_2} \frac{\rho}{m} \\ &= \frac{\exp(-x_P) Q_P}{\widetilde{Q}} n_{O_2} \end{aligned} \tag{2}$$

Substituting Eq. (2) into Eq. (1), we obtain

$$\begin{aligned}
\alpha_P &= \log\left(\frac{Q_P}{n_P}\right) \\
&= \log\left(\frac{\cancel{Q_P}}{\frac{\exp(-x_P)\cancel{Q_P}}{\widetilde{Q}} n_{O_2}}\right) \\
&= \log\left(\frac{\widetilde{Q}}{\exp(-x_P)}\frac{1}{n_{O_2}}\right) \\
&= \log\left(\frac{1}{n_{O_2}}\right) + \log\left(\frac{\widetilde{Q}}{\exp(-x_P)}\right) \\
&= \log\left(\frac{1}{n_{O_2}}\right) + \log\left(\widetilde{Q}\right) + x_P \\
&= \log\left(\frac{\widetilde{Q}}{n_{O_2}}\right) + x_P
\end{aligned}$$

where n_{O_2} is information coming from the *Timescale 0*.

2 Timescale 2

The mass fraction of group $\mathcal{I}_S \in \mathcal{I}_P$ will assume the following formulation

$$Y_{S_P} = \frac{\exp(-x_{S_P})Q_{S_P}}{\widetilde{Q}_P} Y_P \qquad \forall \quad \mathcal{I}_S \in \mathcal{I}_P$$

where x_{S_P} is the single output of the DeepONet for *Timescale 2*.
The following partition functions can also be defined

$$\widetilde{Q}_P = \sum_{I_S \in \mathcal{I}_P} \exp(-x_{S_P})Q_{S_P} \quad , \quad Q_P = \sum_{I_S \in \mathcal{I}_P} Q_{S_P} \qquad \forall \quad \mathcal{I}_P \in \mathcal{I}$$

and

$$Q_{S_P} = \sum_{i \in \mathcal{I}_S} q_i = \sum_{i \in \mathcal{I}_S} g_e g_i \exp\left(-\frac{\epsilon_i}{k_B T}\right) \qquad \forall \quad \mathcal{I}_S \in \mathcal{I}_P$$

The number density of group $\mathcal{I}_S \in \mathcal{I}_P$ can be expressed as follows

$$\begin{aligned}
n_{S_P} &= Y_{S_P}\frac{\rho}{m} \\
&= \frac{\exp(-x_{S_P})Q_{S_P}}{\widetilde{Q}_P} Y_P \frac{\rho}{m} \\
&= \frac{\exp(-x_{S_P})Q_{S_P}}{\widetilde{Q}_P} n_P
\end{aligned} \tag{3}$$

Substituting Eq. (3) into Eq. (1), we obtain

$$
\begin{aligned}
\alpha_{S_P} &= \log\left(\frac{Q_{S_P}}{n_{S_P}}\right) \\
&= \log\left(\frac{\cancel{Q_{S_P}}}{\frac{\exp(-x_{S_P})\cancel{Q_{S_P}}}{\widetilde{Q}_P} n_P}\right) \\
&= \log\left(\frac{\widetilde{Q}_P}{\exp(-x_{S_P})}\frac{1}{n_P}\right) \\
&= \log\left(\frac{1}{n_P}\right) + \log\left(\frac{\widetilde{Q}_P}{\exp(-x_{S_P})}\right) \\
&= \alpha_P - \log\left(Q_P\right) + \log\left(\widetilde{Q}_P\right) + x_{S_P} \\
&= \alpha_P + \log\left(\frac{\widetilde{Q}_P}{Q_P}\right) + x_{S_P}
\end{aligned}
$$